BijliCast: hourly demand intelligence for K-Electric & Pakistan's DISCOs

HackInsight '26 · Team The Coalition · Energy track · All results are generated from gold tables (python -m bijlicast.publish summary); running-cost figures are stated assumptions (docs/business_model.md).

Problem & user

K-Electric's procurement desk must decide how much grid power to plan for each hour of next month. Under Pakistan's new CTBCM market, and with marginal prices up sharply since the Feb 2026 fuel crisis (ISMO), every mis-planned evening-peak megawatt is expensive. Today's simple practice: repeat last year's numbers.

Data (official Pakistani sources + reanalysis)

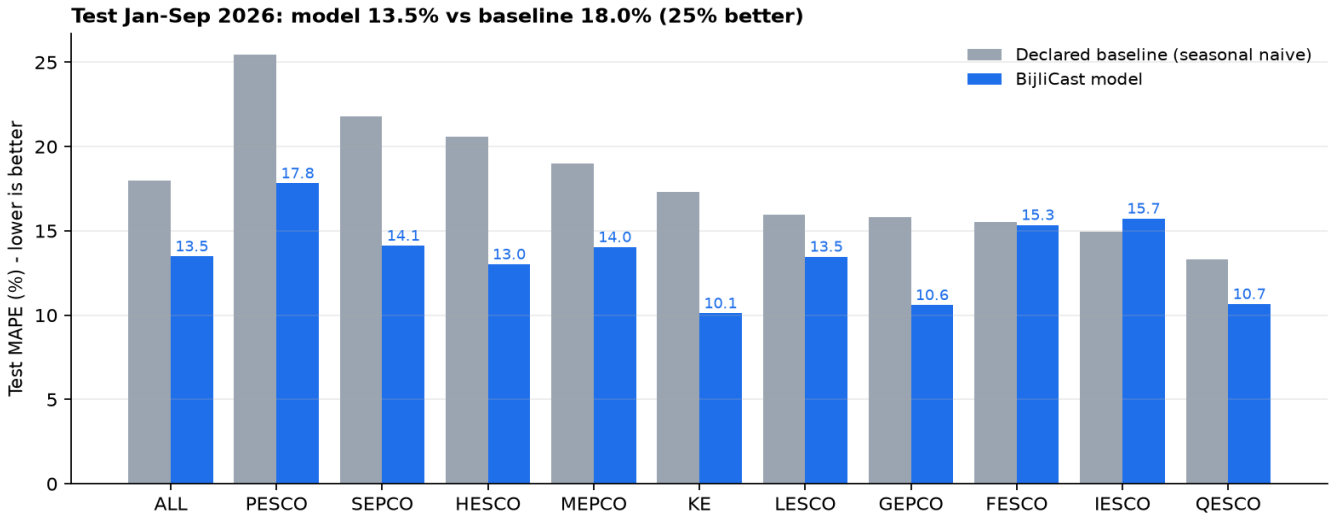
ISMO monthly DISCO/KE workbooks (7,200 hourly-profile rows), ISMO hourly marginal prices (19,728), ERA5 weather for 10 DISCO cities (419,040 hourly rows); NEPRA State of Industry 2024, ISMO IGCEP 2025-35, K-Electric tariff page for citations. Manifest with SHA-256; *make data* rebuilds everything from an empty folder.

Agents & pipeline

Data Quality Agent (pandera + detectors, FIX/FILL/FLAG with reasons): latest status **AMBER** - 7 fixes, 0 fills, 52 flags; it found ISMO months re-uploaded as copies, stale K-Electric values and broken links, and explained Ramadan sehri spikes. **Supervisor agent** + Prefect flow every hour (5 * * * *): quality gate (RED = quarantine, gold keeps last good), drift monitor, retrain only with human approval, forecasts, briefs, golden eval (11 successful runs; full Prefect history incl. 1 failed and 6 cancelled overnight runs in prefect_runs.csv). **Insight Agent (RAG)**: English + Urdu, numbers only from gold rows (IDs stored), citations to document pages, says "I don't know"; golden set 12/12 pass.

Results (test Jan-Sep 2026, evaluated without tuning on it)

Metric	Declared baseline	BijliCast
MAPE, 10 entities x 24 h	18.0%	13.5% (25% better)
K-Electric MAPE / MAE	17.3% / 338 MW	10.1% / 204 MW
Evening peak 18-22 h MAPE	16.9%	12.5%
80% interval coverage	-	75.9%



Impact, limits & business model

Live K-Electric outlook - Oct 2026: mean 2,467 MW, peak 3,082 MW; Nov 2026: mean 2,062 MW, peak 2,513 MW. Priced at ISMO marginal prices, the model cut KE's forecast-mismatch exposure by Rs 83.9 million per average day versus the baseline (gold.error_cost). Weakest: PESCO and summer extremes (under-forecast; coverage below 80%); no gain for IESCO/FESCO - reported, not hidden. Business model: freemium - KE licence + DISCO deployment via the Power Division/donors fund a free public tier with Urdu briefs; running cost about US$20/month compute plus 0.25 FTE. Outputs are planning information, not dispatch or load-shedding instructions (KE helpline 118).